

# Practice Quiz 07C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

*Transfer a limit estimate to a scalar multiple.*

1. Let  $f : \mathbf{R} \rightarrow \mathbf{R}$  and let  $c, L \in \mathbf{R}$ . Suppose  $\lim_{x \rightarrow c} f(x) = L$ . Prove directly from the epsilon-delta definition that

$$\lim_{x \rightarrow c} (-3f(x)) = -3L.$$

**Solution** Let  $\varepsilon > 0$  be arbitrary. Since  $\varepsilon/3 > 0$  and  $\lim_{x \rightarrow c} f(x) = L$ , there is a  $\delta > 0$  such that for every real  $x$ ,

$$0 < |x - c| < \delta \quad \longrightarrow \quad |f(x) - L| < \frac{\varepsilon}{3}.$$

Choose this  $\delta$ . If  $x \in \mathbf{R}$  satisfies  $0 < |x - c| < \delta$ , then

$$|-3f(x) - (-3L)| = 3|f(x) - L| < 3\left(\frac{\varepsilon}{3}\right) = \varepsilon.$$

Since this works for every  $\varepsilon > 0$ , the definition gives  $\lim_{x \rightarrow c} (-3f(x)) = -3L$ . ■